

Tache Only Data

Method

The model is tasked to check whether or not the nasal vowel was normal or clear speech. The architecture used was a Random Forest classifier from sklearn. The model has 1200 estimators each with a depth of 10. The labels used in training are freq f1,amp f1,width f1, freq f2, amp f2, width f2, freq f3, amp f3, width f3, amp p0, freq p0, p0prominence,a1p0,a3p0,vwl amp rms,vwl duration,appendix dur. The timepoint, clarity, and label were dropped for the training set and the only column used to check the prediction was the label column. There is a 70% 30% split on the data for the training and test sets. To check how the model is doing, clacultions of accuract, hamming loss, F1, precision, recall, macro and weighted average, and classification report.

Rseults

Table 1: Classification Report

type	precision	recall	f1-score	support
0 (clear)	66%	76%	71%	140
1 (normal)	76%	67%	71%	162
accuracy			71%	302
macro avg	71%	71%	71%	302
weighted avg	71%	71%	71%	302

When we look at Table 1, we see that the model levels off at around 71%. It is interesting to see that for the classification of normal speech, the precision of the model is 76%. It is hard to find why the model stabalizes at 71% but this is something to look further. I also calculated the hamming loss and accuracy. The accuracy came out to 70.86%. The hamming loss is a metric specifically for multi-class models, like this random forest. According to Scikit learn, "The Hamming loss is the fraction of labels that are incorrectly predicted." Given that, the model had a hamming loss of around 0.29 which is not desirable since we would want the loss to be as near 0

When we look at Figure 1, we see that all of the features play a role but out of the columns in the csv file, the vowel amplitude rms feature played the most into the decision tree prediction. The MDI stands for "Mean Decreased in Impurity" which, according to

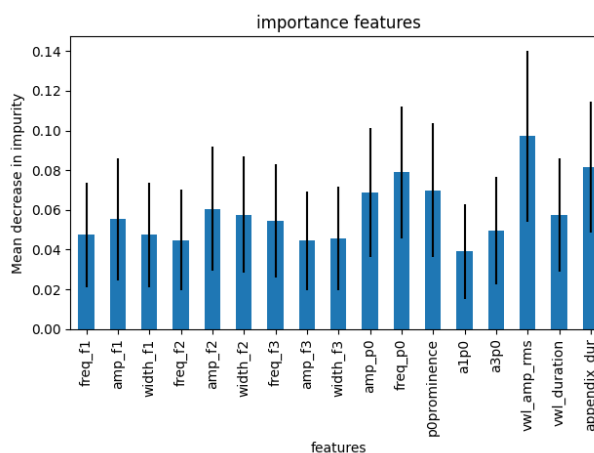


Figure 1: Feature importances using MDI

sklearn themselves, is computed by the mean and standard deviation of accumulation of the impurity decrease within each tree. MDI calculates the total reduction of node impurity across all of the trees.

There were no one feature that solely decides the random forest that does the prediction. Using the MDI, shows how each feature contributes in the decision of the prediction. The drawback with using MDI is that it can be biased towards numerical features or columns with many unique values.

Figure 2 shows the importance of the features using permutations on the full model. In this case, the features are shuffled n times, twice in Figure 2 and the model then predicts on the shuffled data. This approach on contrast to the MDI importance, does not take into account of biases and check each feature and their contribution. We can see the two features that are helping the model make the predictions, being the vowel amplitude rms and the appendix duration.

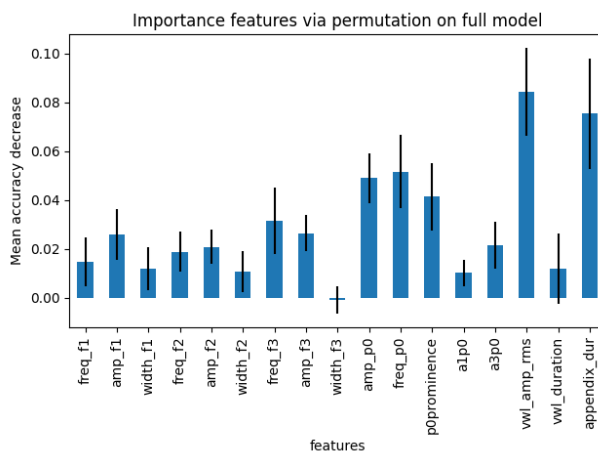


Figure 2: Feature importances using permutation

Tache and Lecture Data

Method

The model is tasked to check whether or not the nasal vowel was normal or clear speech. The architecture used was a Random Forest classifier from sklearn. The model has 1000 estimators each with a depth of 30. The labels used in training are freq f1, amp f1, width f1, freq f2, amp f2, width f2, freq f3, amp f3, width f3, amp p0, freq p0, p0prominence, a1p0, a3p0, vwl amp rms, vwl duration, appendix dur. The search query, speaker, task, clarity, ND, timepoint, word, vowel, and target columns were dropped for the training set and the only column used to check the prediction was the label column. There is a 80% 20% split on the data for the training and test sets. To check how the model is doing, calculations of accuracy on the validation set, precision, recall, F1-score, macro average, weighted average, and out of the box (OOB) score.

Results

The full report is shown in table 2. We can see the increase in all values with across the board wavering at least around 80%. This is a big gain in the classification regards compared to just the lecture data due to the increase of data, as we can see that for

Table 2: Classification Report

type	precision	recall	f1-score	support
0 (clear)	80%	82%	81%	394
1 (normal)	82%	81%	81%	411
accuracy			81%	805
macro avg	81%	81%	81%	805
weighted avg	81%	81%	81%	805

the normal has increased to 411 entries to support and in the clear has increased to 394 entries.

Table 3 provides the val accuracy and the OOB score. According to the Scikit-learn official website, the OOB score (or error from their website) is as follows:

the average error for each $z_i = (x_i y_i)$ calculated using predictions from the trees that do not contain z_i in their respective bootstrap sample.

Table 3: Other Report

type	result
Validation accuracy	81.12%
OOB score	80.37%

The validation accuracy is just how well the model did on the set that was not seen in the training set. Figure 3 shows what feature plays a factor in the random forest.

In this case, it is the vowel amplitude rms. We can see that the appendix duration is up next with the second feature that the random forest model will look at. Interestingly, the features of nasality does not play a role in the decision.

In Figure 4, we can see how much the model relies on the vowel amplitude rms and the appendix duration where they are almost the same at 0.14. In both figures, we can see a sharp drop in the third feature of the actual vowel duration, at around 0.08. We can see in figure 4 that the features of nasality other than P0 are low, not reaching 0.02. The amplitude and frequency of P0 is far lower in the chain of features but it is interesting that it comes before the formants of the vowel and a1-p0

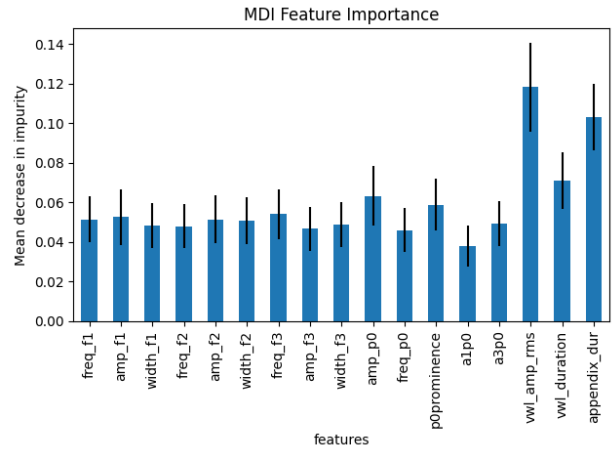


Figure 3: Feature importances using MDI

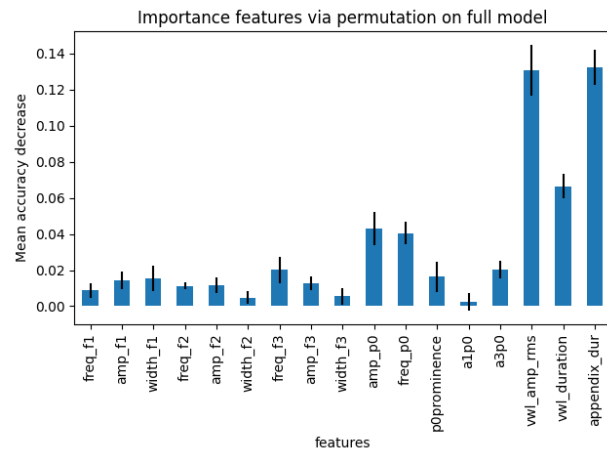


Figure 4: Feature importances using permutation